

Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	1 Mark

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only a limited number of concurrent users on the local Flask server.	Performance may decrease when multiple users access the application simultaneously.	High
2	Uses a static agricultural dataset with limited crop varieties.	Predictions are limited to the crops available in the dataset.	Medium
3	Runs only as a local web application without cloud deployment.	Users cannot access the application remotely.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports only local users.	Deploy the application on cloud platforms such as Render, AWS, or Azure with load balancing.
2	Data Storage	Dataset stored as a CSV file.	Integrate a relational database such as MySQL or PostgreSQL for efficient data management.
3	Performance	Prediction model runs on a single Flask server.	Optimize the model, use caching, and deploy with Gunicorn/Nginx for improved performance.
4	Security	Basic local application security.	Implement user authentication, HTTPS, secure API handling, and input validation.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud deployment and database integration	1–2 Months	Improve accessibility, scalability, and data management.

Phase 2	Weather API integration for real-time environmental data	2–3 Months	Increase prediction accuracy with live weather information.
Phase 3	Mobile application for Android and iOS	3–5 Months	Enable farmers to access crop recommendations from anywhere.
Phase 4	AI-powered fertilizer recommendation, irrigation scheduling, and multilingual support	6+ Months	Enhance smart farming capabilities and improve usability for a wider audience.